

EBRAHIM ALHADDAD

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PROFESSIONAL SUMMARY

Software engineer focused on complex systems, with experience spanning embedded systems, performance analysis, backend services, and LLM-powered applications. Combines engineering experience with growing expertise in statistical and machine learning modeling and applied analytics.

SKILLS

Key Skills & Tools:

AI & ML: LangChain | LangGraph | Vector Databases | RAG | LLM APIs | scikit-learn | Recommender Systems | LLMOps

Data & Analytics: Pandas | Polars | SQL | EDA | Feature Engineering | Statistical Modeling | Experimentation

Backend & Cloud: Python | FastAPI | Docker | AWS (CDK, ECS, ECR, Cognito, Amplify, Bedrock) | OpenSearch (Elasticsearch)

Systems & Performance: C | C++ | Embedded Systems | Multithreaded Programming | Performance Analysis

Certification: IBM AI Engineering Professional Certificate

Languages (Spoken): English, Arabic

EDUCATION

Master of Science in Business Analytics, Rady School of Management
University of California, San Diego 06/2026
San Diego, CA

Honors: Crown Prince International Scholarship (2025 - 2026), Rady Fellowship (2025 - 2026)

Bachelor of Science in Electrical Engineering, Viterbi School of Engineering
University of Southern California 12/2019
Los Angeles, CA

Honors: Crown Prince International Scholarship (2013 - 2019), USC Dean's List (2015 - 2016)

EXPERIENCE

Rarible 05/2025 - 07/2025
Software Engineer — contractor Remote

- Collaborated with stakeholders to translate creative user flows into technical architecture and development plans
- Designed a full-stack AI-driven agentic NFT generation platform using FastAPI, Next.js and LangGraph
- Built scalable backend using Redis, DynamoDB, SQL and Claude / Bedrock APIs enabling chat, code execution and NFT generation
- Configured CI/CD and cloud deployment flows using CDK on AWS for production-ready infrastructure

Array Innovation (Fast-Growing AI Startup) 05/2024 - 05/2025
Tech Lead/AI Engineer — Data Science & AI Team Bahrain

- Led projects as startup scaled from 10 to 100+ employees; managed 9 engineers and partnered with C-suite on hiring and strategy
- Acted as interim product owner, defining roadmaps and demos. Engaged with clients to translate requirements into technical design
- Led development of LLM-powered systems integrating FastAPI services, vector search and automated evaluation pipelines
- Built data pipelines and retrieval systems supporting keyword extraction, vector search, RAG and AI workflows
- Boosted search accuracy and retrieval performance by 20% through embedding optimization, fuzzy matching, and TF-IDF tuning

Apple 08/2021 - 02/2023
Software Engineer — Core Media IMG Team San Diego, CA

- Collaborated on a large, highly parallelized C codebase at the system level, maintaining core media APIs across all Apple devices
- Facilitated discussions among experts to align on solutions for high-priority bugs, strengthening collaboration
- Brought technical insights from debugging sessions to managers' meetings, shaping prioritization and resource allocation
- Debugged complex memory, power, latency and multithreading issues across Unix systems; triaged over 9,000 bugs
- Developed automation tools for log analysis, enabling trend extraction and issue tracking to speed up resolution of critical bugs

Qualcomm 05/2020 - 08/2021
Performance Engineer — (Machine Learning / Computer Vision) R&D Team San Diego, CA

- Produced stakeholder-ready reports distilling complex performance metrics for senior management
- Presented analysis findings to influence design decisions for new product planning and future SoC architectures
- Prepared and balanced datasets for computer vision models
- Brought up RCMs, configured lab equipment and developed scripts to automate experiments and data acquisition
- Measured SoC power and bandwidth under different parameters and resource configurations (CPUs, DSPs, GPUs, Cache)
- Conducted function-level analysis on computer vision pipelines and computed fine-grained performance KPIs

PROJECTS

Recommender Model for Research Papers — Recommendation Systems 12/2025

- Developed Top-N recommendation model on 1.1M+ author-paper interactions using SVD, NMF and LightFM Warp
- Engineered features (popularity, activity decay, concept similarity using sparse n-gram embedding) to improve implicit feedback
- Improved Hit-Rate@K from 36% (SVD) to 72%, achieving 2x lift over matrix factorization baseline

Algorithmic Trading Personal Project — Software Engineering 08/2023

- Collaborated with a financial research group to implement an automated, adaptive derivatives trading strategy
- Designed real-time software using results from financial analyses and developed a Python project utilizing QuantConnect API